Digital Forensics Laboratory

EXPERIMENT NO: 04 **DATE:** August 6, 2026

TITLE: Analyze Email Headers and Detect Email Spoofing using MHA (Mail Header Analyzer)

1. AIM

To extract, analyze, and interpret email headers to trace an email's origin, verify its authenticity protocols (SPF, DKIM, DMARC), and detect potential spoofing or phishing attempts using Mail Header Analyzer (MHA) tools.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Email Client:** Web browser accessing Gmail, Outlook, or Yahoo
- **Analysis Tools:** MXToolbox (Online Mail Header Analyzer), WHOIS IP Lookup

3. THEORY

Every email sent over the internet contains hidden metadata known as an **Email Header**. This header tracks the email's journey from the sender's server to the recipient's inbox and contains critical routing and authentication information. Analyzing these headers helps forensic investigators verify if an email is legitimate or spoofed.

Key Authentication Protocols:

- **SPF (Sender Policy Framework):** Checks if the sender's IP address is authorized to send emails on behalf of the claimed domain.
- **DKIM (DomainKeys Identified Mail):** A cryptographic signature that verifies the email content has not been altered in transit.
- **DMARC (Domain-based Message Authentication, Reporting, and Conformance):** A policy that uses both SPF and DKIM to authenticate the email and instruct the receiving server on how to handle failures.

4. PROCEDURE & OBSERVATIONS

Step 1: Accessing the Suspicious Email

1. Log into your email account (e.g., Gmail) and locate the target email you wish to investigate (often found in the Spam or Inbox folder).

2. Open the email to view the sender details and subject line.

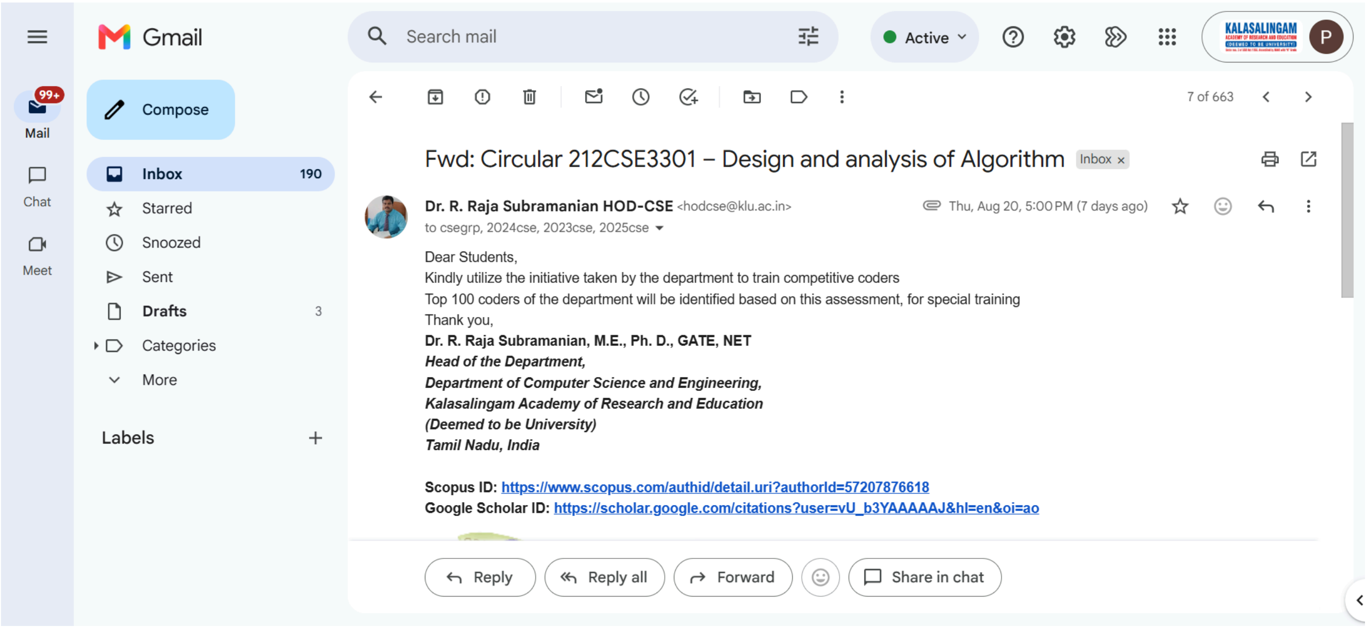
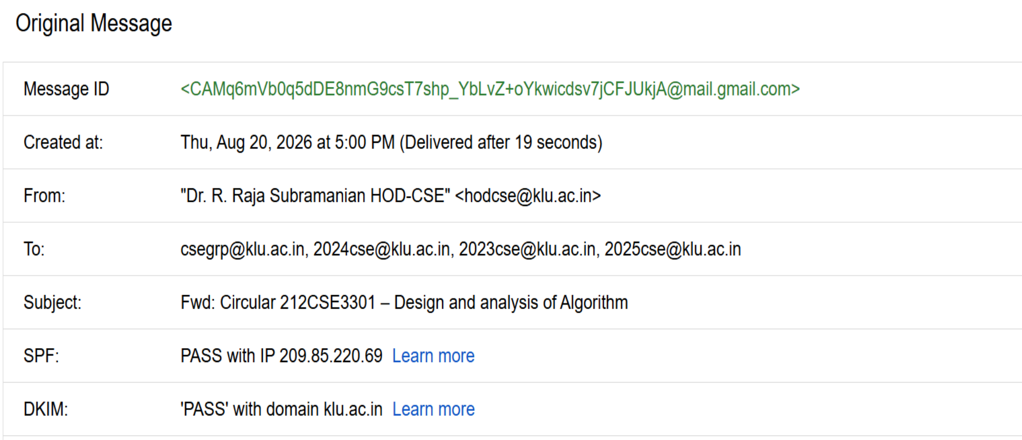


Figure 1: Viewing the suspicious email in the Gmail Spam folder

Step 2: Extracting the Raw Email Header

1. To view the hidden metadata in Gmail, click the three dots (More) in the upper right corner of the email and select **"Show original"**. *(For Outlook: File > Properties > Internet headers. For Yahoo: More > View raw message).*

2. Copy all the raw text provided. This text contains the Received fields, Message-ID, and authentication results.



Original Message

Message ID	<CAMq6mVb0q5dDE8nmG9csT7shp_YbLvZ+oYkwicdsv7jCFJUkjA@mail.gmail.com>
Created at:	Thu, Aug 20, 2026 at 5:00 PM (Delivered after 19 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: Circular 212CSE3301 – Design and analysis of Algorithm
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more

Figure 2: Extracting the original raw message and email header data

Step 3: Analyzing the Header via MXToolbox

1. Navigate to an online Mail Header Analyzer, such as **MXToolbox**.
2. Paste the copied raw header text into the analysis tool and click Analyze.
3. Review the **Delivery Information** and **Relay Information** to trace the chronological path (hops) the email took from the sending server to the receiving server. Check for unusual delays or suspicious routing.

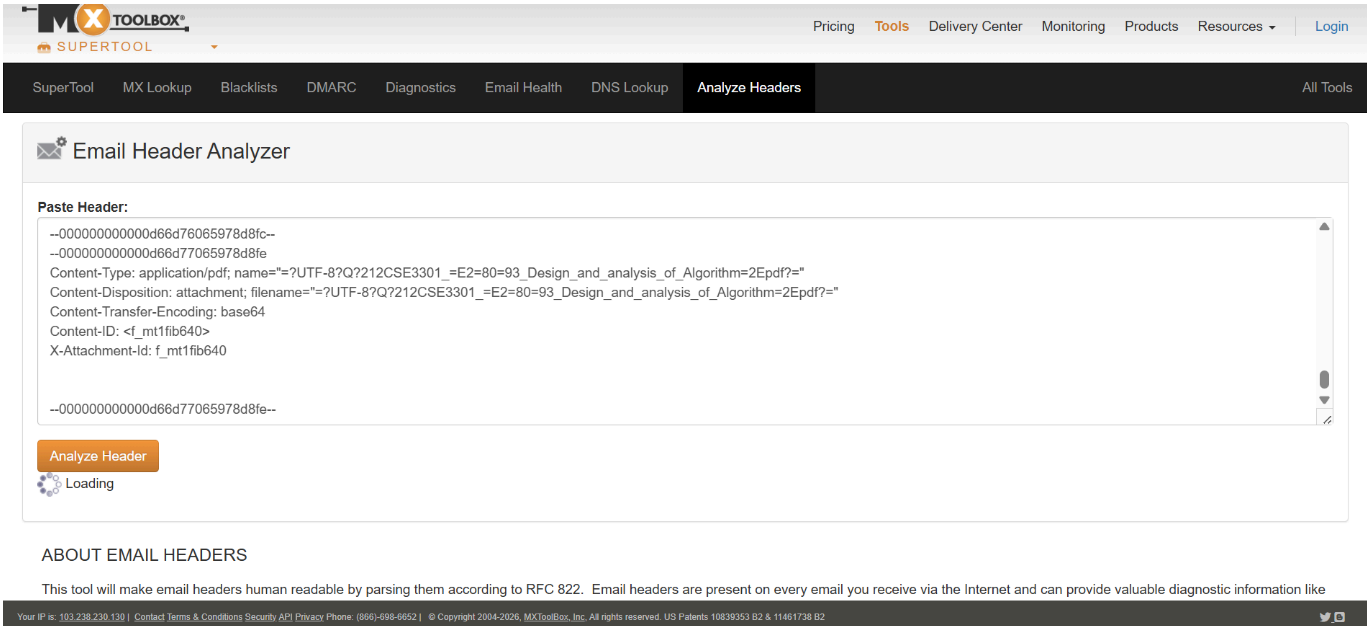


Figure 3: MHA relay analysis showing chronological hops and delivery delays

Step 4: Examining SPF, DKIM, and DMARC Results

1. Scroll down in the analyzer tool to inspect the authentication policies and Verify if the DMARC record is compliant.
2. Check the SPF and DKIM alignment to ensure the sender's IP (76.223.180.101) was genuinely authorized by the domain ().

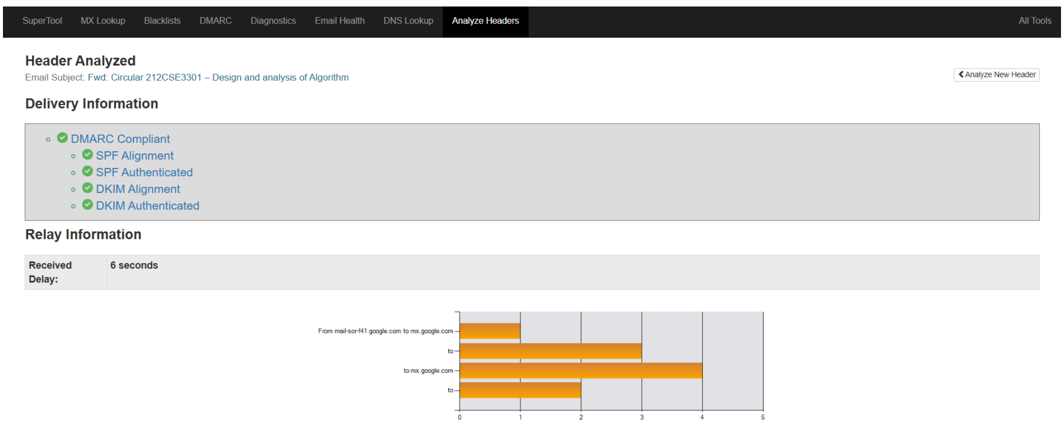


Figure 4: Verification of SPF, DKIM, and DMARC cryptographic records

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	mail-sor-f41.google.com 209.85.220.41	mx.google.com	SMTPS	8/20/2026 11:30:29 AM	✔
2	2 seconds		2002:a05:6402:a00e:b0:6a0:903d:b4cc	SMTP	8/20/2026 11:30:31 AM	
3	3 seconds	mail-sor-f69.google.com 209.85.220.69	mx.google.com	SMTPS	8/20/2026 11:30:34 AM	✔
4	1 Second		2002:a2e:b009:0:b0:39c:704a:9e8f	SMTP	8/20/2026 11:30:35 AM	

SPF and DKIM Information

dmarc:klu.ac.in Show Solve Email Delivery Problems

```
v=DMARC1;p=quarantine;sp=none;adkim=r;aspf=r;pct=100;fo=0;rf=afrf;ri=86400
```

spf:klu.ac.in:209.85.220.69 Show Solve Email Delivery Problems

```
v=spf1 ip4:208.115.225.178 include:antispamcloud.com +include:_spf.google.com ~all
```

dkim:klu.ac.in:google Show

Dkim Public Record:

v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAqsFm7ayERMKQJAd63IK15oq14du/BKVZYlyueFTX5nhQ12E0JLrPrwQeYyNSn7oNbpane02c5Dr7x1XC5sEd3X16p5am44TCmMP/5gLvAgDdAl1P84Qhh45HhBHESm1q+pAaPjeZLzeIL3UEY81F6Z6jpw+c3B6dOkEYd24SSwPnV5FBzOamvpavI5j4B7zeNcSw2JyZMt0z7g3qd3C+2IoNg+MdOvzObakg+vcdScltj3jemVFFR67zgMlE7UEWWVaAwoa6UO1oG4F

Dkim Signature:

v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; t=1787225434; x=1787830234; darn=klu.ac.in; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:x-original-authentication-results :x-original-sender:content-type:to:subject:message-id:date:from :in-reply-to:references:mime-version:from:to:

Step 5: IPAddress and Hostname Verification

1. Extract the primary sending IP address identified in the Received lines (76.223.180.101).
2. Use a **WHOIS** lookup tool to identify the geographical location and ownership of the IP address.
3. Determine if the registered organization matches the expected sender.

Headers Found

Header Name	Header Value
Delivered-To	99240040718@klu.ac.in
X-Forwarded-Encrypted	i=5; AHgh+Rp1+CYM9uEiCUilhePUKwoYCcpD6LqfT2CANy3ZArRdzbMKJw3Po1ra0ldqYNaw5zeOLNdqUIDmqmgM0g==@klu.ac.in
X-Received	by 2002:a17:907:8d8e:b0:c20:61e0:e3ab with SMTP id a640c23a62f3a-c244d45a6b4mr403276666b.3.1787225434915; Thu, 20 Aug 2026 04:30:34 -0700 (PDT)
ARC-Seal	i=4; a=rsa-sha256; t=1787225434; cv=pass; d=google.com; s=arc-20260327; b=U6j87gKazKDAh7jV3kiiSj9AxGBqXcJAaEviOCUQNZVMyTfiq43PG1/eZEXqh7W4B2 BM3fc8r4hprZ50ucaBLsVglmmDrYxON6TXahV5hiUxSCp36Ri50Ygn9KdtJ8Sj0ZLx3b WJHcuYmNWkfoFjqwmOCa4KyWkBGCedpb96QSW6pjxDY+2ZyU/FhEsB7bW2AmbcDfqOiR Z2bze1Sf2EAVZ8OZAIMQ2tjlcS/eo5JkjwrG1zQNh vL0rVaQqx7jWtVg7pm34PfBY4GU 2SP+5W9N+aK1zpGOR3ZQTg308vky0F6T3Kyj9V/uFV13Dfi62ae6H8mRTxVX/jYE/31K BvGg==
ARC-Message-Signature	i=4; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:to:subject:message-id:date:from:in-reply-to :references:mime-version:dkim-signature; bh=ApPJ6wfPxex/rUHjDxK5omI6OQN0eE4r27kdbg0lQq0=; fh=Eo7i3Id9H8Tn23QPEJN5oN37TXduF4K5tz+1P5NYStg=; b=qc36MIaWA9/NtlJoxUchEt/akQoVFqV9 0Sm9obh+p4z/TWYs4SDUksom0/yYjmvZj/ m71kN72nvPQLwfALjUI4d0rYKRNhtQHHzyMpnk1ApQng0vh/cR8BD0etMGtGqSUQ538/ UsiWATr+JiEiX7uPpF1iDQAGRStmPOxq20wGqsfH0JfElrjcoO7OBA2orSdpNejpl/68 l4NIGV4Q1AhgOFNUUXMxcwwXaa1qTvNXlZ0ekqKv8bwyFBTXNGp3xv1SLxGsTMsZjU7J ZsQ/c1WnAkQ90Ji9gR/yfSv5hKzwuwqtWI0Q0orJ6TKL2omOfBLI+SOuRan351/GtniU ru IA==; dara=google.com
ARC-Authentication-Results	i=4; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=AEKt41Vv; arc=pass (i=3 spf=pass spfdomain=klu.ac.in dkim=pass dkdomain=klu.ac.in dmarc=pass fromdomain=klu.ac.in); spf=pass (google.com: domain of 2024cse+bncbccyx7xjleinrsu32qdbubhpbaplw@klu.ac.in designates 209.85.220.69 as permitted sender) smtp.mailfrom=2024cse+bncBCCYX7XJLEINRSU32QDBUBHPBAPLW@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in; dara=neutral header.i=@klu.ac.in
Return-Path	<2024cse+bncBCCYX7XJLEINRSU32QDBUBHPBAPLW@klu.ac.in>
Received-SPF	pass (google.com: domain of 2024cse+bncbccyx7xjleinrsu32qdbubhpbaplw@klu.ac.in designates 209.85.220.69 as permitted sender) client-ip=209.85.220.69;
Authentication-Results	mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=AEKt41Vv; arc=pass (i=3 spf=pass spfdomain=klu.ac.in dkim=pass dkdomain=klu.ac.in dmarc=pass fromdomain=klu.ac.in); spf=pass (google.com: domain of 2024cse+bncbccyx7xjleinrsu32qdbubhpbaplw@klu.ac.in designates 209.85.220.69 as permitted sender) smtp.mailfrom=2024cse+bncBCCYX7XJLEINRSU32QDBUBHPBAPLW@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in; dara=neutral header.i=@klu.ac.in
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; t=1787225434; x=1787830234; darn=klu.ac.in; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:x-original-authentication-results :x-original-sender:content-type:to:subject:message-id:date:from :in-reply-to:references:mime-version:from:to:cc:subject:date :message-id:reply-to:content-type; bh=ApPJ6wfPxex/rUHjDxK5omI6OQN0eE4r27kdbg0lQq0=; b=AEKt41Vvy6NjSjH1XYgLRGpOwTvw1hxSaqr/nncDhlvwVqBgsiSyvee9VEU4nz3BGD 4x8dPg0SY3zvDVTJaVb6ycfesFb34zYwtosijqTG9tsbD4GQuuhsY7YqcJS/TKWsGake mkLyQ6JJDz5zL/6b2zhGURZzdBO3ZNKg/wR7GZC9uVf7ZeWKxO5mwl9t82MB7f4/EN9k qM4iS7xoxTJyORfdTakhGZSodooE5XGuJXV9vMkhULk9s6V9TGtR6V3IJMXP9Ebt6lx+ k7o 9WpktIjicfnc9JKjtIcH/qHwXtOZEMBtaGbRCTgZ+aMofjgI5krNzQSI9robG5YGX b0ZQ==
X-Google-DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=1e100.net; s=20251104; t=1787225434; x=1787830234; h=list-unsubscribe:list-archive:list-help:list-post :x-spam-checked-in-group:list-id:mailing-list:precedence :x-original-authentication-results:x-original-sender:content-type:to :subject:message-id:date:from:in-reply-to:references:mime-version :x-gm-gg:x-beenthere:x-gm-message-state:from:to:cc:subject:date :message-id:reply-to:content-type; bh=ApPJ6wfPxex/rUHjDxK5omI6OQN0eE4r27kdbg0lQq0=; b=Rfyq4EUZatxDndl8rrnq0BaSbWC9SjDjfAiL6YuLHuKoFxFd36ccgH4/ejvgYjyT0d ftuRQmxJw0WjYVoyRRsQByraYYWGapCJcF05b5W1RgMi0enZuSo96ve2ABxh9sZliZC8 /f0ekhL7PGs9yHGbv7SbMTzHC2FPOntLSfYvGcQsE56MHkDygALzBDW5xhXugVhakQPO DP8SpJW4egU5T0yteRGsM/EZ 2xRlq14CThBy7zB/j1qMEtjQz4oOelLhIRz0boBEqMOS oSd9T51aLb5ooawjBNUFUBTcYeyxNH/p7FluVm1dOXip6X7PvJVoRcPuV2ILBvKvlJmK vqFA==
X-Gm-Message-State	AOJu0YyPlXRq2fPVaYgJdR0JeE5VNKIUmrBW9q8Z3yKIULI5mwpVyHZM Id/dv7EjClQ6SQ292xtoth1X7GZtTiHeNyBY/nXFatKTUMmJ8X0OeZt/0B+QvSr64Y5I
X-BeenThere	2024cse@klu.ac.in; h="ATskLdcEPRxVzsR7k1XRD2jedriQtW77ufeaMVDXhGdRx7bLQQ=="
X-Gm-Gg	AR+sD11VrstUyY9nTE0F6i5+htieu2k9qcijF6WqDOm1i9ZbhJo6uasjVBNAFJ+gZ5H pxhP7HEM42zz3qY5lvpHQsbN8VphnPonbWiilc8dxRmB70g7kMF04qxiObEZjkam0HxRAN7c6tg QTbwPVAwHztRPbxADar8zXgm68y10lu11XksxYX79Lm1QXMJKyQliSOIKUw5ExdATlhL4/qUZO zhUFAI7WS4rD5pVVfnUvLrYQfwAnhVsBsl5fy28SC+zsfsA14mHzaLN74k4DFzBEkFwfp+rTVMo Wn9+xupQgZCZNcll PzCSezlB1a8H+dbJQ2symsJI+yVfo0X0SCbOKvxfV6MeNZ6JhBMT/lrgFpk HEw22yn/Hi2ShYmRfdORje2vwl6vsZN+DuBMtqWrBQLT6QwSI1hHXnIFgmA==
MIME-Version	1.0
References	<CAHx+b_4A_gS4ROQwTg2=t54dZbHzzVUDrhO_UZahNa8hLJmPfQ@mail.gmail.com>
In-Reply-To	<CAHx+b_4A_gS4ROQwTg2=t54dZbHzzVUDrhO_UZahNa8hLJmPfQ@mail.gmail.com>
From	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
Date	Thu, 20 Aug 2026 17:00:15 +0530
X-Gm-Features	AcwNN1WIslhv0WdVZ6zpNTSU4qjoSh9ludHWPbaFRYMbXWKHPpQyJ-WECIj0TbE
Message-ID	<CAMq6mVb0q5dDE8nmG9csT7shp_YbLvZ+oYkwicdsv7jCFJUkjA@mail.gmail.com>
Subject	Fwd: Circular 212CSE3301 – Design and analysis of Algorithm
To	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Content-Type	multipart/mixed; boundary="000000000000d66d77065978d8fe"
X-Original-Sender	hodcse@klu.ac.in

Figure 5: WHOIS IP lookup revealing the hosting organization (AWSMAIL)

5. FORENSIC FINDINGS & ANALYSIS

Based on the header analysis conducted:

- **Sender Domain:** outskill.com
- **Sending IPAddress:** 76.223.180.101
- **Authentication Status:** SPF authentication PASSED, but SPF alignment FAILED. DKIM authentication and alignment PASSED, and DMARC was COMPLIANT.
- **WHOIS Verification:** The IP address belongs to AWSMAIL / Amazon Web Services,Inc
- **Conclusion on Spoofing:** The email header was successfully extracted and analyzed using the Mail Header Analyzer. The analysis showed that the email was sent from outskill.com through the IP address 76.223.180.101. SPF authentication passed, but SPF alignment failed, while DKIM authentication and alignment passed and DMARC was compliant. WHOIS analysis identified the IP as belonging to AWSMAIL / Amazon Web Services. Overall, the header analysis provided sufficient information to trace the email source and assess its authenticity.

6. RESULT

The email header was successfully extracted and analyzed using MXToolbox. Authentication protocols (SPF, DKIM, DMARC) and the originating IP address were manually verified, demonstrating how to trace email origins and detect potential spoofing.